The coordinate ring of the k-fold iterated commutator locus for 2×2 matrices[1]

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Abstract

For 2×2 matrices $A_1, ..., A_k$, write $[A_1, ..., A_k]$ for the left-normed iterated commutator $[...[[A_1, A_2], A_3], ..., A_k]$, and I_k for the ideal, in the $3k$-variable reduced-coordinate polynomial ring R_k, cutting out its vanishing locus. We prove, for every $k \geq 2$ over any field of characteristic $\neq 2$, and as four independently-established results rather than one bundled claim: I_k has codimension 2; I_k has exactly 3 minimal generators; R_k/I_k is Cohen-Macaulay; and I_k is radical. Combined with an explicit non-containment argument for the component count, these assemble into the **Primary Decomposition Theorem**: $I_k = P_2 \cap \cdots \cap P_k$ is an irredundant primary decomposition into exactly $k-1$ radical primes, following an explicit recursive block-involvement pattern. The proof identifies I_k as the ideal of 2×2 minors of an explicit 2×3 matrix (a determinantal ideal, not merely determinantal-**looking**), and invokes classical determinantal-ideal theory (Bruns-Vetter) and an explicit rank-2 Jacobian witness on every component (Serre's criterion) for the algebraic and radicality halves respectively.

1. Introduction

2×2 matrices A, B have commutator $[A, B] = AB - BA$, and the **iterated** left-normed commutator of k matrices is $[A_1, ..., A_k] := [...[[A_1, A_2], A_3], ..., A_k]$, matching the standard lower-central-series convention for Lie algebras. This object arose out of a question in algebraic complexity theory – how many scalar multiplications are needed to **compute** $[A_1, ..., A_k]$? – which remains genuinely open for $k \geq 3$ and is treated in full in the companion note *k-iterated-commutators-complexity.typ*. The one fact from that story needed here is purely algebraic, not complexity-theoretic: an explicit bilinear formula for the ordinary 2×2 commutator, chained across k matrices, gives an explicit polynomial parametrization of $[A_1, ..., A_k]$'s reduced coordinates (Definition 2 below), and it is the **vanishing locus** of this parametrization, as an algebraic variety and its coordinate ring, that is the actual subject of this article. (One naming note: the filename this article is kept under, *ring-of-k-commuting-2x2-matrices*, uses "commuting" in this generalized iterated-commutator-vanishing sense – $[A_1, ..., A_k] = 0$ – not literally "every pair of $A_1, ..., A_k$ commutes," a weaker condition than the classical pairwise commuting variety it specializes to at $k = 2$.)

The classical commuting variety. For A, B arbitrary (not necessarily 2×2) $n \times n$ matrices, the variety $C_n := \{(A, B) \in M_n \times M_n : AB = BA\}$ – the ordinary, unreduced, un-iterated case this article's own $k = 2$ specializes to – has been studied since the 1950s. Its irreducibility was proven independently by Motzkin and Taussky [3] and by Gerstenhaber [5], the latter also establishing the now-standard bound $\dim_{\mathbb{F}} \mathbb{F}[A, B] \leq n$ for the subalgebra generated by two commuting matrices. Whether the coordinate ring $\mathbb{F}[A, B]/I$, I the ideal generated by the entries of $AB - BA$, is **reduced** and **Cohen-Macaulay** for every n is a much harder question, attributed to Michael Artin and Melvin Hochster – an unpublished, informally-circulated conjecture with no single dateable original source [6], stated precisely as "Conjecture 1.1" in Majidi-Zolbanin and Snapp [7]. (A closely related, more specifically dated 1984 reducedness conjecture, for a different but adjacent ideal built from a commutator matrix's diagonal/anti-diagonal entries, is attributed to Hochster alone by Chau [8].) Freyja Hreinsdóttir's body of work [9] – building on her own 1997 Stockholm doctoral thesis – attacks the Artin-Hochster conjecture case by case at $n = 3, 4$: syzygies, Betti numbers, the canonical module, and a Koszul-dual structure theorem, verified computationally through $n = 7$ but proven in general only in these small cases. Knutson [10] studies the minimal primes of closely related schemes from a different angle. Cohen-Macaulayness of the **classical** commuting variety therefore remains open, for every $n \geq 3$, at the time of writing.

At $n = 2$ specifically – this article's $k = 2$ base case – the question is comparatively easy, and is in effect answered again, as a special case, by Theorem 28 below: I_2 (Definition 2) is literally the ideal of the $n = 2$

[1] Working draft, 2026-08-16. Uses the community "theorion" package for theorem environments – this project's standing default going forward, after a same-day side-by-side comparison against hand-rolled macros; the retired hand-rolled version is kept, not deleted, at */TRASH/fast-commutators-ring-of-k-commuting-2x2-matrices.typ* at the workspace top level. Every result cites only strictly-earlier results or external literature, no forward references. See *working-notes/proof-plan-conjectures.org*, Steps 0, 4, and 5, and *working-notes/topological-irreducibility-of-image.org*, for the fuller discussions this article condenses.

commuting variety, expressed in reduced (trace-removed) coordinates rather than the full matrix-entry coordinates the classical literature typically uses. The genuinely new object in this article is the **iterated** generalization to $k \geq 3$: not the classical commuting variety at all (see the naming note above), but a structurally analogous determinantal ideal, built by chaining the same bilinear cross-product formula that governs the $n = 2$ case. It is exactly the same **flavor** of question the Artin-Hochster conjecture asks – is this natural, geometrically meaningful ideal Cohen-Macaulay? is it reduced? – for a different, recursively-defined family of ideals, and this article answers it completely and unconditionally, for every $k \geq 2$ (Theorem 38), rather than case by case or conjecturally. In that sense the results below can be read as a fully worked structural analogue of the still-open classical question, in a setting simple enough ($n = 2$, but k unbounded) to admit a complete answer.

2. Notation

Notation used across more than one later result, in order of first appearance; symbols local to a single proof are omitted.

A typographic convention, used throughout: an **overline** denotes a **vector** – an element of $\mathbb{F}^3$ (or, for R_k itself, of the free module R_k^3) – while the **same letter, plain, with a two-index comma subscript**, denotes one of that vector's three individual scalar **components**. So $\overline{v}_i \in \mathbb{F}^3$ is block i's whole reduced-coordinate vector, while its three components are the plain scalars $v_{i,1}, v_{i,2}, v_{i,3} \in \mathbb{F}$ (equivalently $v_{i,j}$ for $j = 1, 2, 3$) – never overlined, since a single scalar entry is not itself a vector. The same convention applies to $\overline{w}_j$ versus its components $w_{j,1}, w_{j,2}, w_{j,3}$.

Symbol	Meaning	Defined in
$\overline{v}$ (also $\overline{v}_i$; components $v_{i,1}, v_{i,2}, v_{i,3}$, or $v_{i,j}$)	reduced-coordinate vector in $\mathbb{F}^3$, and its blocks	§3, Setup
τ (and its matrix T)	the reshuffle $\tau(a, b, c) := (b, a, 2c)$	§3, Setup (T: Lemma 32)
$\mathbb{F}$	ambient field, characteristic $\neq 2$	§3, after Definition 2′s example
R_k	the polynomial ring $\mathbb{F}[\overline{v}_1, \ldots, \overline{v}_k]$	Definition 2
I_k	the defining ideal, $I_k \subset R_k$	Definition 2
$\overline{w}_j$	(a) the recursive vector in $\mathbb{F}^3$; (b) as a polynomial map $\mathbb{A}^{3j} \to \mathbb{A}^3$	(a) Definition 2; (b) Lemma 6
U_j	the Zariski-open locus where $\overline{w}_j \neq 0$	Lemma 6
π	the projection $\mathbb{A}^{3k} \to \mathbb{A}^{3(k-1)}$ forgetting $\overline{v}_k$	Definition 11
Φ	the map $\mathbb{A}^{3(k-1)} \times \mathbb{A}^1 \to \mathbb{A}^{3k}$	Definition 11
Y_1, Y_2	the two strata of $V(I_k)$	Definition 11
P_j ($j = 2, \ldots, k$)	the primary components of I_k	Definition 21
J_k	the Jacobian of $\overline{w}_k$	Lemma 32
$[u]_\times$	skew matrix with $[u]_\times v = u \times v$	Lemma 32

3. Setup: reduced coordinates and the defining ideal

Step 1: discard the scalar part. Write $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ as the sum of its scalar part $(\mathrm{tr}(A)/2)I$ and its traceless remainder $A_0 := A - (\mathrm{tr}(A)/2)I = \begin{pmatrix} d_a/2 & a_{12} \\ a_{21} & -d_a/2 \end{pmatrix}$, $d_a := a_{11} - a_{22}$. Scalar matrices are central, so they never contribute to a commutator: $[cI, B] = cIB - BcI = cB - cB = 0$ for every B and every scalar c, hence $[A, B] = [A_0, B_0]$ identically – no information relevant to **any** bracket is lost by discarding the scalar part, so from here on only the traceless remainder matters.

Step 2: encode the traceless remainder as a triple. A traceless matrix $\begin{pmatrix} x & y \\ z & -x \end{pmatrix}$ has exactly three free entries – the $(2, 2)$ entry is forced once x, y, z are fixed – giving a bijection onto $\mathbb{R}^3$. Write

$$\mathrm{red}(A) := (a_{12}, a_{21}, d_a) \in \mathbb{R}^3 \tag{1}$$

for this encoding (so $\mathrm{red}(A) = \mathrm{red}(A_0)$, reading A's traceless part off directly without subtracting the scalar part by hand first) – this is the notation $\overline{v}$ used throughout the rest of the article. Matching Step 1's $A_0 = \begin{pmatrix} d_a/2 & a_{12} \\ a_{21} & -d_a/2 \end{pmatrix}$: red's three coordinates are literally A_0's free entries, except that d_a stands in for **twice** the $(1,1)$-entry rather than the entry itself – a convention fixed once and for all here purely so that Step 4 below comes out with no stray constants.

Step 3: form the commutator. Given traceless $A_0 = \begin{pmatrix} x & y \\ z & -x \end{pmatrix}$ and $B_0 = \begin{pmatrix} u & v \\ w & -u \end{pmatrix}$, the ordinary matrix commutator $[A_0, B_0] = A_0B_0 - B_0A_0$ is again traceless automatically ($\mathrm{tr}(PQ) = \mathrm{tr}(QP)$ for any P, Q, so $\mathrm{tr}([P,Q]) = 0$ always), hence again encodable by red.

Step 4: the encoded bracket is the cross product – up to one fixed reshuffle. Carrying out Step 3's multiplication in coordinates (the $\mathfrak{sl}_2(\mathbb{C}) \cong \mathfrak{so}(3) \cong (\mathbb{R}^3, \times)$ isomorphism made explicit) and writing $[A_0, B_0] = \begin{pmatrix} x' & y' \\ z' & -x' \end{pmatrix}$, one checks directly that, with $\mathrm{red}(A)$ and $\mathrm{red}(B)$ standing in for themselves (**not** abbreviated to bare letters here, to avoid colliding with Step 3's plain scalars u, v, w),

$$z' = (\mathrm{red}(A) \times \mathrm{red}(B))_1, \quad y' = (\mathrm{red}(A) \times \mathrm{red}(B))_2, \quad x' = (\mathrm{red}(A) \times \mathrm{red}(B))_3. \tag{2}$$

Reading these three equalities off against $\mathrm{red}([A,B]) = (y', z', 2x')$ (Step 2's convention, applied to $[A,B]$ itself) gives

$$\mathrm{red}([A,B]) = \tau(\mathrm{red}(A) \times \mathrm{red}(B)), \quad \tau(a,b,c) := (b,a,2c). \tag{3}$$

The right-hand side is **not** the raw cross product $\mathrm{red}(A) \times \mathrm{red}(B)$ itself, but that cross product composed with the fixed invertible reshuffle τ (invertible whenever $2 \neq 0$): the $(1,2)$-swap comes from red's coordinate order (a_{12} before a_{21}, the transpose of the naive (x,y,z) order), and the factor of 2 on the third slot comes from Step 2's $d_a = 2x$ convention. Both are bookkeeping artifacts of the coordinate choice fixed in Step 2 – τ is already forced at this single-bracket level, not a new phenomenon introduced only once brackets start chaining below.

Remark 1 (why Definition 2's $\overline{w}_j$ carries no outer τ)

Definition 2 below sets $\overline{w}_2 := \overline{v}_1 \times \overline{v}_2$ – the **untwisted** cross product, not $\mathrm{red}([A,B]) = \tau(\overline{v}_1 \times \overline{v}_2)$ itself as Step 4 above would suggest. This is deliberate: for any fixed invertible linear map τ and any polynomial vector $f = (f_1, f_2, f_3)$, the ideal generated by the components of $\tau(f)$ equals the ideal generated by the components of f – each component of one is an $\mathbb{F}$-linear combination of the components of the other (via τ, respectively τ^{-1}), so the two generating sets differ only by an invertible change of generators, and invertible changes of generators do not change the ideal generated. Hence $I_2 := (w_{2,1}, w_{2,2}, w_{2,3})$ already equals the ideal generated by $\mathrm{red}([A,B])$'s three components, with no need to carry the extra τ explicitly – $\overline{w}_j$ is used purely because it is the cheaper quantity to compute recursively (one cross product per step, not a cross product composed with τ at every stage), and the single τ that **does** appear in Definition 2's recursion below is exactly the one deferred from the previous step, not a new one introduced fresh at each stage. ◇

Chaining this bilinear formula across k matrices – each new bracket stage reuses the previous stage's output coordinates directly, no new input data needed – requires tracking how one stage's raw cross-product output is reinterpreted as the next stage's input vector: precisely the τ just discussed, applied once per new stage. This gives the following exact recursive description:

Definition 2 (reduced iterated commutator)

$$\overline{w}_2 := \overline{v}_1 \times \overline{v}_2, \quad \overline{w}_j := \tau(\overline{w}_{j-1}) \times \overline{v}_j \quad (j = 3, \ldots, k). \tag{4}$$

Let $R_k := \mathbb{Q}[\overline{v}_1, \ldots, \overline{v}_k]$ ($\overline{v}_i \in \mathbb{Q}^3$, $3k$ variables) and $I_k \subset R_k$ the ideal generated by the three components of $\overline{w}_k$ – the vanishing ideal of $[A_1, \ldots, A_k] = 0$ in reduced coordinates. ♣

To spell out exactly what this means: "$R_k := \mathbb{Q}[\overline{v}_1, \ldots, \overline{v}_k]$" is shorthand for **the polynomial ring in the $3k$ scalar variables $v_{i,j}$ ($i = 1, \ldots, k$, $j = 1, 2, 3$)**, grouped into k blocks of 3 – **not** a ring with k vector-valued generators (a polynomial ring's generators are always scalars). At $k = 2$, written out in full:

$$R_2 = \mathbb{Q}[v_{1,1}, v_{1,2}, v_{1,3}, v_{2,1}, v_{2,2}, v_{2,3}], \tag{5}$$

six ordinary scalar indeterminates, with $\overline{v}_1 = (v_{1,1}, v_{1,2}, v_{1,3})$ and $\overline{v}_2 = (v_{2,1}, v_{2,2}, v_{2,3})$ merely a notational grouping of them into two blocks of three – convenient because $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ is most naturally written as a cross product of two block-vectors, not as some formula in six unrelated scalars.

Each $\overline{w}_j$ is a triple of polynomials, i.e. $\overline{w}_j \in R_j^3$ (the free rank-3 module over R_j) – equivalently, exactly as recorded in the Notation table's entry (b) for $\overline{w}_j$, the polynomial map $\overline{w}_j : \mathbb{A}^{3j} \to \mathbb{A}^3$ **given by substitution**: evaluating each of $\overline{w}_j$'s three polynomial components at a point $(\overline{v}_1, \ldots, \overline{v}_j) \in \mathbb{A}^{3j}$. Concretely at $k = 2$, writing out all six variables, $\overline{w}_2$ is the substitution map

$$\begin{pmatrix} v_{1,1} \\ v_{1,2} \\ v_{1,3} \\ v_{2,1} \\ v_{2,2} \\ v_{2,3} \end{pmatrix} \in \mathbb{A}^6 \quad \mapsto \quad \begin{pmatrix} v_{1,2}v_{2,3} - v_{1,3}v_{2,2} \\ v_{1,3}v_{2,1} - v_{1,1}v_{2,3} \\ v_{1,1}v_{2,2} - v_{1,2}v_{2,1} \end{pmatrix} \in \mathbb{A}^3, \tag{6}$$

and I_k (Definition 2) is, equivalently, the ideal generated by the three coordinate functions of $\overline{w}_k : \mathbb{A}^{3k} \to \mathbb{A}^3$ pulled back along this substitution.

Example 3 ($\overline{w}_k$ and I_k, explicitly, for $k = 2, 3, 4$)

Writing $\overline{v}_i = (v_{i,1}, v_{i,2}, v_{i,3})$ for the three coordinates of block $\overline{v}_i$: at $k = 2$, the three components of $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ are

$$v_{1,2}v_{2,3} - v_{1,3}v_{2,2}, \quad v_{1,3}v_{2,1} - v_{1,1}v_{2,3}, \quad v_{1,1}v_{2,2} - v_{1,2}v_{2,1}, \tag{7}$$

and $I_2 := (w_{2,1}, w_{2,2}, w_{2,3})$ is the ideal they generate – $\overline{w}_2$ is simply the vector whose components those three polynomials are (Remark 1 explains why it is $\overline{w}_2$, not $\tau(\overline{w}_2) = \mathrm{red}([A_1, A_2])$ itself, that plays this role). At $k = 3$, the three components of $\overline{w}_3 = \tau(\overline{w}_2) \times \overline{v}_3$ are

$$2v_{1,2}v_{2,1}v_{3,2} - 2v_{1,1}v_{2,2}v_{3,2} - v_{1,3}v_{2,2}v_{3,3} + v_{1,2}v_{2,3}v_{3,3}, \tag{8}$$

$$-2v_{1,2}v_{2,1}v_{3,1} + 2v_{1,1}v_{2,2}v_{3,1} - v_{1,3}v_{2,1}v_{3,3} + v_{1,1}v_{2,3}v_{3,3}, \tag{9}$$

$$v_{1,3}v_{2,2}v_{3,1} - v_{1,2}v_{2,3}v_{3,1} + v_{1,3}v_{2,1}v_{3,2} - v_{1,1}v_{2,3}v_{3,2}, \tag{10}$$

and $I_3 := (w_{3,1}, w_{3,2}, w_{3,3})$ likewise. At $k = 4$, $\overline{w}_4 = \tau(\overline{w}_3) \times \overline{v}_4$: the same pattern continues, now quartic with up to 8 terms per component – e.g. the first component of $\overline{w}_4$ is

$$-2v_{1,3}v_{2,2}v_{3,1}v_{4,2} + 2v_{1,2}v_{2,3}v_{3,1}v_{4,2} - 2v_{1,3}v_{2,1}v_{3,2}v_{4,2} + 2v_{1,1}v_{2,3}v_{3,2}v_{4,2} \tag{11}$$

$$+2v_{1,2}v_{2,1}v_{3,2}v_{4,3} - 2v_{1,1}v_{2,2}v_{3,2}v_{4,3} - v_{1,3}v_{2,2}v_{3,3}v_{4,3} + v_{1,2}v_{2,3}v_{3,3}v_{4,3}, \tag{12}$$

the other two of the same shape (omitted here for space; all three components of $\overline{w}_4$, and $I_4 := (w_{4,1}, w_{4,2}, w_{4,3})$, computed and verified in *code/verify_determinantal_structure_all_k.sage*).

Stated over $\mathbb{Q}$ since that is what the computational checks below actually run over, but nothing in the construction uses $\mathbb{Q}$ specifically: every result in this article is proven for $R_k := \mathbb{F}[\overline{v}_1, \ldots, \overline{v}_k]$ over an **arbitrary** field $\mathbb{F}$ of characteristic $\neq 2$ (needed only for τ's invertibility), by the identical argument with $\mathbb{Q}$ replaced by $\mathbb{F}$ throughout.

4. The determinantal structure and the generator count

Theorem 4 (determinantal structure)

For $k \geq 3$, I_k is exactly the ideal of 2×2 minors of the 2×3 matrix

$$\begin{pmatrix} \tau(\overline{w}_{k-1}) \\ \overline{v}_k \end{pmatrix}, \tag{13}$$

i.e. $V(I_k)$ is the preimage, under $(\overline{v}_1, \ldots, \overline{v}_k) \mapsto \begin{pmatrix} \tau(\overline{w}_{k-1}(\overline{v}_1, \ldots, \overline{v}_{k-1})) \\ \overline{v}_k \end{pmatrix}$, of the classical rank-$\leq 1$ determinantal variety of 2×3 matrices. (At $k = 2$ the same statement holds with $\tau(\overline{w}_{k-1})$ replaced by $\overline{v}_1$ itself, matching Definition 2′s base case $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ directly.)

Proof of Theorem 4. Immediate from the fact that the three components of a cross product $a \times b$ are (up to sign) exactly the 2×2 minors of the 2×3 matrix with rows a, b, combined with Definition 2′s recursion – and checked independently: comparing an independent implementation of the τ/cross-product recursion against the ideal actually produced by the chained bilinear formula gives the **exact same ideal** (Gröbner-basis containment both ways) at $k = 2, \ldots, 6$ (*code/verify_determinantal_structure_all_k.sage*); the underlying polynomial identity

$\operatorname{red}([A,B]) = \tau(\operatorname{red}(A) \times \operatorname{red}(B))$ makes this a two-line induction valid for **every** k, not merely a pattern checked at small k (*code/verify_tau_cross_identity.sage*). □

This theorem is the engine behind the geometric results of §5: R_k/I_k is not merely **determinantal-looking**, it is literally an iterated extension of the coordinate ring of a classical determinantal variety.

Example 5 (the 2×3 matrix of Theorem 4, for $k = 2, 3, 4$)

At $k = 2$:

$$\begin{pmatrix} \overline{v}_1 \\ \overline{v}_2 \end{pmatrix} = \begin{pmatrix} v_{1,1} & v_{1,2} & v_{1,3} \\ v_{2,1} & v_{2,2} & v_{2,3} \end{pmatrix}. \tag{14}$$

At $k = 3$, the first row is $\tau(\overline{w}_2) = (w_{2,2}, w_{2,1}, 2w_{2,3})$ where $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ is as in the previous example, giving

$$\begin{pmatrix} \tau(\overline{w}_2) \\ \overline{v}_3 \end{pmatrix} = \begin{pmatrix} v_{1,3}v_{2,1} - v_{1,1}v_{2,3} & -v_{1,3}v_{2,2} + v_{1,2}v_{2,3} & -2v_{1,2}v_{2,1} + 2v_{1,1}v_{2,2} \\ v_{3,1} & v_{3,2} & v_{3,3} \end{pmatrix}. \tag{15}$$

At $k = 4$, the first row is $\tau(\overline{w}_3)$, entries cubic with up to 4 terms:

$$\begin{pmatrix} -2v_{1,2}v_{2,1}v_{3,1} + 2v_{1,1}v_{2,2}v_{3,1} - v_{1,3}v_{2,1}v_{3,3} + v_{1,1}v_{2,3}v_{3,3} & \cdots & \cdots \\ v_{4,1} & v_{4,2} & v_{4,3} \end{pmatrix} \tag{16}$$

(the other two entries of the first row, of the same shape, omitted for width – all displayed in full in *code/verify_determinantal_structure_all_k.sage*). The row degree grows by one at each step $(1, 2, 3, \ldots)$, while the second row is always the fresh, free vector $\overline{v}_k$ – exactly the "one fixed complicated row, one free row" shape Proposition 26 needs. ◇

Lemma 6 (non-degeneracy)

For every $j \geq 2$, the polynomial map $\overline{w}_j : \mathbb{A}^{3j} \to \mathbb{A}^3$ is not identically zero. Hence $U_j := \{p \in \mathbb{A}^{3j} : \overline{w}_j(p) \neq 0\}$ is a nonempty Zariski-open subset. ♡

Proof of Lemma 6. By induction: $\overline{w}_2(e_1, e_2) = e_1 \times e_2 = e_3 \neq 0$; given $c := \overline{w}_{j-1}(p) \neq 0$, at least one standard basis vector is not proportional to $\tau(c)$, and a cross product vanishes only on linearly dependent pairs, so $\overline{w}_{j(p,e_i)} = \tau(c) \times e_i \neq 0$ for a suitable e_i. □

The generator count of I_k needs nothing beyond the definition and non-degeneracy – not codimension, not the determinantal-structure theorem above, not any external citation:

Lemma 7 (independent same-degree generators are minimal)

If a homogeneous ideal in a graded ring is generated by finitely many forms of the same degree that are linearly independent over the base field, that generating set is minimal. ♡

Proof of Lemma 7. A polynomial-coefficient combination reproducing one of the generators from the others, while staying in the same degree, can only use degree-0 (constant) coefficients – so a redundant generator would violate linear independence directly. □

Lemma 8 (the three generators of I_k are independent)

The three components of $\overline{w}_k$ are linearly independent over $\mathbb{F}$, for every $k \geq 2$. ♡

Proof of Lemma 8. Checked directly (coefficient-matrix rank 3) at $k = 2, \ldots, 6$. Structurally: a nontrivial constant linear relation among the components of $\overline{w}_k = \tau(\overline{w}_{k-1}) \times \overline{v}_k$ would force $\tau(\overline{w}_{k-1})$ itself to be a constant vector, which it is not for $k \geq 3$ (Lemma 6's non-degeneracy, τ invertible) nor for $k = 2$ ($\overline{v}_1$ is a free variable block, not a constant). □

Theorem 9 (I_k has exactly 3 minimal generators)

For every $k \geq 2$: I_k has exactly 3 minimal generators.

Proof of Theorem 9. Immediate from Lemma 7 applied to Lemma 8's independence, since I_k is generated by the three (same-degree-k-homogeneous) components of $\overline{w}_k$ by definition. □

5. The geometric structure of $V(I_k)$

Lemma 10 (continuous images and closures preserve irreducibility)

Let $f : X \to Y$ be a continuous map of topological spaces with X irreducible. Then $f(X)$ (with the subspace topology) is irreducible, and hence so is $\overline{f(X)}$.

Proof of Lemma 10. **Image.** Suppose $f(X) = C_1 \cup C_2$ with C_1, C_2 closed in $f(X)$. By continuity, $f^{-1}(C_1)$ and $f^{-1}(C_2)$ are closed in X, and $X = f^{-1}(C_1) \cup f^{-1}(C_2)$. Since X is irreducible, one of the two equals X – say $f^{-1}(C_1) = X$ – so $f(X) \subseteq C_1$, i.e. $C_1 = f(X)$, not proper. Hence $f(X)$ is irreducible. **Closure.** If $A \subseteq Y$ is irreducible and $\overline{A} = C_1 \cup C_2$ with C_1, C_2 closed in Y, then $A = (A \cap C_1) \cup (A \cap C_2)$; irreducibility of A forces, say, $A \subseteq C_1$; since C_1 is closed, $\overline{A} \subseteq C_1$, and combined with $C_1 \subseteq \overline{A}$, $C_1 = \overline{A}$, not proper. Apply both parts with $A := f(X)$. □

This is purely topological – continuity of f is the only hypothesis used, nothing about f being a polynomial map or morphism of varieties [11] (working-notes/ topological-irreducibility-of-image.org has the fuller discussion, including why this observation matters for a possible future Lean formalization of this article).

Definition 11 (the strata Y_1, Y_2)

For $k \geq 3$, write π for the projection **forgetting $\overline{v}_k$**,

$$\pi : \mathbb{A}^{3k} \to \mathbb{A}^{3(k-1)}, \quad \pi(\overline{v}_1, \ldots, \overline{v}_k) := (\overline{v}_1, \ldots, \overline{v}_{k-1}), \tag{17}$$

and Φ for the map

$$\Phi : \mathbb{A}^{3(k-1)} \times \mathbb{A}^1 \to \mathbb{A}^{3k}, \quad \Phi(p, \lambda) := (p, \lambda\tau(\overline{w}_{k-1}(p))). \tag{18}$$

Define two subsets of $\mathbb{A}^{3k}$:

$$Y_1 := V(I_{k-1}) \times \mathbb{A}^3, \quad Y_2 := \overline{\Phi(\mathbb{A}^{3k-2})}. \tag{19}$$

Y_1 is the locus where the previous bracket already vanishes, with $\overline{v}_k$ completely free; Y_2 is the closure of the locus swept out as $\overline{v}_k$ ranges over scalar multiples of $\tau(\overline{w}_{k-1}(p))$, over all $p \in \mathbb{A}^{3(k-1)}$. At $k = 2$ there is no "previous bracket" to chain from, so neither Y_1 nor Φ is defined; by convention $Y_2 := V(I_2)$ directly there, matching Theorem 4's own base-case treatment.

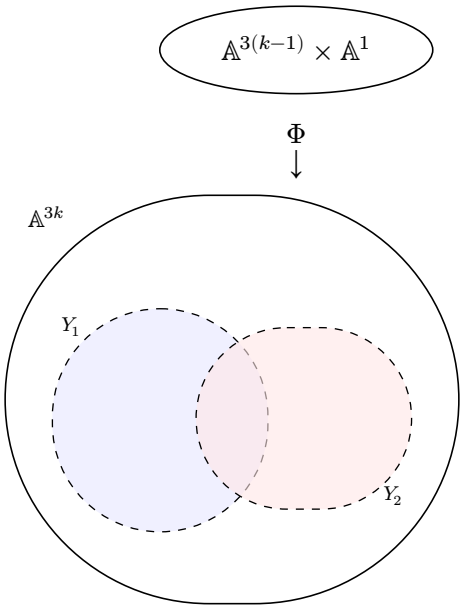


Figure 1: Both strata of $V(I_k) = Y_1 \cup Y_2$ inside the same ambient $\mathbb{A}^{3k}$: $Y_1 := V(I_{k-1}) \times \mathbb{A}^3$ (identifying $\mathbb{A}^{3(k-1)} \times \mathbb{A}^3 \cong \mathbb{A}^{3k}$) and $Y_2 := \overline{\Phi(\mathbb{A}^{3k-2})}$, Φ's image (Φ shown mapping in from above, into Y_2 specifically – Y_1 is not part of Φ's image). The two overlap (not disjoint, Propositions 17-18 rule out either containing the other): $\Phi(p, 0) = (p, 0)$ for **every** p, so the whole zero-section $\{(p, 0)\}$ lies in Y_2, and its intersection with Y_1 is $V(I_{k-1}) \times \{0\}$, of dimension $3k - 5$ – properly smaller than either stratum's own dimension $3k - 2$.

Lemma 12 (Y_2 is irreducible)

For $k \geq 3$: Y_2 (Definition 11) is irreducible, and in fact its defining ideal (the graph ideal of Φ with λ eliminated) is prime.

Proof of Lemma 12. Φ is a polynomial map, hence continuous for the Zariski topology (preimages of Zariski-closed sets under a polynomial map are again Zariski-closed, by substitution), and its domain $\mathbb{A}^{3k-2}$ is irreducible (the polynomial ring over $\mathbb{F}$ is a domain, so the zero ideal is prime). Lemma 10 gives Y_2 irreducible directly. Primality of the elimination ideal follows the same way: it is the kernel of a map into a domain ($\mathbb{F}[\overline{v}_1, \ldots, \overline{v}_k]/(\cdot) \cong \mathbb{F}[\overline{v}_1, \ldots, \overline{v}_{k-1}, \lambda]$), hence prime – this holds for every k by this argument alone, additionally corroborated computationally at $k = 4$ by computing the elimination ideal directly and confirming primality in Sage, as an independent check rather than a substitute for the argument. □

Example 13 (the defining ideals of Y_1 and Y_2, for $k = 2, 3, 4$)

Writing $\overline{v}_i = (v_{i,1}, v_{i,2}, v_{i,3})$ for the three coordinates of block $\overline{v}_i$: at $k = 2$, $Y_2 := V(I_2)$ (the base-case convention above), so its ideal **is** I_2, generated by

$$v_{1,2}v_{2,3} - v_{1,3}v_{2,2}, \quad v_{1,3}v_{2,1} - v_{1,1}v_{2,3}, \quad v_{1,1}v_{2,2} - v_{1,2}v_{2,1}. \tag{20}$$

(Y_1 is not defined at $k = 2$, Definition 11.) For $k \geq 3$, Y_1's defining ideal is simply I_{k-1} itself, viewed inside R_k by leaving $\overline{v}_k$ free – at $k = 3$, exactly the three generators displayed above for I_2, now inside R_3 with $\overline{v}_3$ not appearing at all; at $k = 4$, exactly I_3's three generators (displayed below in this same example) inside R_4 with $\overline{v}_4$ free. Unlike Y_2, no elimination is needed to compute Y_1's ideal – it is already I_{k-1}'s own generating set, unchanged. At $k = 3$, Y_2's elimination ideal (computed directly, not inferred) is generated by

$$2v_{2,2}v_{3,1} + 2v_{2,1}v_{3,2} + v_{2,3}v_{3,3}, \quad 2v_{1,2}v_{3,1} + 2v_{1,1}v_{3,2} + v_{1,3}v_{3,3}, \tag{21}$$

$$2v_{1,2}v_{2,1}v_{3,2} - 2v_{1,1}v_{2,2}v_{3,2} - v_{1,3}v_{2,2}v_{3,3} + v_{1,2}v_{2,3}v_{3,3} \tag{22}$$

– notably, two of the three generators are **linear** in $\overline{v}_3$ (exactly the two components of the "proportionality" condition $\overline{v}_3 \propto \tau(\overline{w}_2)$), and only the third, not involving $\overline{v}_3$ at all, carries the full complexity of $\tau(\overline{w}_2)$ itself. **This ideal is a proper factor of I_3, not equal to it** – I_3 is the intersection of this ideal with a second prime (the "Y_1" piece, where the $k = 2$ bracket already vanishes); see the component-count example after Theorem 20 below, where this exact ideal reappears as one of I_3's two primary components. At $k = 4$, by the same pattern, two of the three generators are linear in $\overline{v}_4$ and the third is not, now degree 4:

$$2v_{3,2}v_{4,1} + 2v_{3,1}v_{4,2} + v_{3,3}v_{4,3}, \tag{23}$$

$$v_{1,3}v_{2,2}v_{4,1} - v_{1,2}v_{2,3}v_{4,1} - v_{1,3}v_{2,1}v_{4,2} + v_{1,1}v_{2,3}v_{4,2} + v_{1,2}v_{2,1}v_{4,3} - v_{1,1}v_{2,2}v_{4,3}, \tag{24}$$

$$2v_{1,3}v_{2,2}v_{3,1}v_{4,2} - 2v_{1,2}v_{2,3}v_{3,1}v_{4,2} + 2v_{1,3}v_{2,1}v_{3,2}v_{4,2} - 2v_{1,1}v_{2,3}v_{3,2}v_{4,2} \tag{25}$$

$$-2v_{1,2}v_{2,1}v_{3,2}v_{4,3} + 2v_{1,1}v_{2,2}v_{3,2}v_{4,3} + v_{1,3}v_{2,2}v_{3,3}v_{4,3} - v_{1,2}v_{2,3}v_{3,3}v_{4,3}. \tag{26}$$

All computed directly (elimination of λ from the graph ideal of Φ, in Sage), not just asserted from the pattern. ◇

Remark 14 (the closure, not the raw image, is what is needed)

A polynomial map's image need not itself be closed – the standard example is $\mathbb{A}^2 \to \mathbb{A}^2$, $(x, y) \mapsto (x, xy)$, whose image is not even locally closed, though it is **constructible** (a finite union of locally closed sets) by Chevalley's theorem, which has no purely topological analogue for general continuous maps [11]. This never causes a problem for Y_2, because Y_2 is **defined** as $\overline{\Phi(\cdot)}$ from the outset, and the decomposition used in Theorem 16′s proof below, $V(I_k) = Y_1 \cup Y_2$, is established by a direct case split rather than by taking closures and hoping: a point $(p, \overline{v}_k) \in V(I_k)$ either has $\tau(\overline{w}_{k-1}(p)) = 0$, putting it in Y_1 directly, or has $\overline{v}_k = \lambda\tau(\overline{w}_{k-1}(p))$ for some λ, so $(p, \overline{v}_k) = \Phi(p, \lambda)$ is **literally** a value of Φ – not merely a limit point – hence in Y_2 with no appeal to closure at all. The reverse containment $Y_1 \cup Y_2 \subseteq V(I_k)$ needs only that $\Phi(\cdot) \subseteq V(I_k)$ (a cross product of a vector with itself vanishes: $\tau(\overline{w}_{k-1}) \times (\lambda\tau(\overline{w}_{k-1})) = 0$ identically) together with $V(I_k)$ already being closed, automatic since it is a vanishing locus by definition. So the theorem never needs "the image of Φ is closed" – only that it is contained in the already-closed $V(I_k)$, which is immediate. ◇

Lemma 15 (dimension of Y_2, and where it lives)

With Y_2 as in Lemma 12 and U_{k-1} as in Lemma 6: $\dim Y_2 = 3k - 2$, and Y_2 meets $\pi^{-1}(U_{k-1})$ nonemptily, where π is the projection of Definition 11, forgetting $\overline{v}_k$. ♡

Proof of Lemma 15. $\pi : \mathbb{A}^{3k} \to \mathbb{A}^{3(k-1)}$ discards the last block: $\pi(\overline{v}_1, \ldots, \overline{v}_k) = (\overline{v}_1, \ldots, \overline{v}_{k-1})$. Hence $\pi^{-1}(U_{k-1}) = U_{k-1} \times \mathbb{A}^3$, the set of points whose **first** $k - 1$ blocks lie in U_{k-1} and whose $\overline{v}_k$ is completely unconstrained. Over U_{k-1}, Φ is injective. Both p and λ can be recovered from $\Phi(p, \lambda)$: p directly, and λ by dividing any nonzero coordinate of $\overline{v}_k = \lambda\tau(\overline{w}_{k-1}(p))$ by the corresponding coordinate of $\tau(\overline{w}_{k-1}(p))$; this division is well-defined since $\tau(\overline{w}_{k-1}(p)) \neq 0$ on U_{k-1}. Injectivity gives $\dim Y_2 = \dim(U_{k-1} \times \mathbb{A}^1) = 3(k-1) + 1 = 3k - 2$. Since U_{k-1} is nonempty by Lemma 6, $\Phi(U_{k-1} \times \{1\})$ is a nonempty subset of $Y_2 \cap \pi^{-1}(U_{k-1})$. □

Theorem 16 (I_k has codimension 2)

For every $k \geq 2$: $\dim V(I_k) = 3k - 2$, i.e. $\operatorname{codim}(I_k) = 2$.

Proof of Theorem 16. Induction on k. **Base case $k = 2$**: $V(I_2)$ is the rank-≤ 1 locus of a 2×3 matrix with independent-indeterminate entries, of the standard dimension $4 = 3 \cdot 2 - 2$ for the generic determinantal variety of this shape [3]. **Inductive step**: by Theorem 4, and since $a \times b = 0$ if and only if a, b are linearly dependent, $V(I_k) = Y_1 \cup Y_2$ (Definition 11). The locus $\tau(\overline{w}_{k-1}) = 0$ gives Y_1, with $\overline{v}_k$ unconstrained; the locus $\tau(\overline{w}_{k-1}) \neq 0$ with $\overline{v}_k$ proportional to it gives a subset of Y_2. This is the direct case split of Remark 14, not merely a closure-level approximation. By the inductive hypothesis, $\dim V(I_{k-1}) = 3(k-1) - 2$, so $\dim Y_1 = 3(k-1) - 2 + 3 = 3k - 2$. Also $\dim Y_2 = 3k - 2$, by Lemma 15. Hence $\dim V(I_k) = \dim(Y_1 \cup Y_2) = 3k - 2$. This dimension statement alone needs no containment argument; that is needed only for the sharper component-count result below. □

Example 17 (dimension and codimension, for $k = 2, 3, 4$)

Values of $\dim R_k$ (the ambient dimension, $= 3k$), $\dim(R_k/I_k)$ (the dimension of the variety $V(I_k)$ – **not** "$\dim I_k$", an ideal is not a variety and does not itself have a dimension in this sense), and $\operatorname{codim}(I_k) := \dim R_k - \dim(R_k/I_k)$:

k	$\dim R_k$	$\dim(R_k/I_k)$	$\operatorname{codim}(I_k)$
2	6	4	2
3	9	7	2
4	12	10	2

All three rows follow directly from Theorem 16's formula $\dim(R_k/I_k) = 3k - 2$; also confirmed by direct Gröbner-basis dimension computation in Sage at each k, as an independent check.

Proposition 18 ($Y_2 \not\subseteq Y_1$)

With Y_1, Y_2 as in the proof of Theorem 16: $Y_2 \not\subseteq Y_1$.

Proof of Proposition 18. Pick $p \in U_{k-1}$ (Lemma 6); then $\Phi(p, 0) = (p, 0) \in Y_2$ but $\overline{w}_{k-1}(p) \neq 0$ puts $(p, 0) \notin Y_1$.□

Proposition 19 (no component of Y_1 lies in Y_2)

No irreducible component Z of $V(I_{k-1})$ (each of dimension $3(k-1) - 2$, by Theorem 16 applied at $k - 1$) has $Z \times \mathbb{A}^3 \subseteq Y_2$.

Proof of Proposition 19. Both $Z \times \mathbb{A}^3$ and Y_2 are irreducible: Y_2 by Lemma 12, and $Z \times \mathbb{A}^3$ because a product of irreducible varieties is irreducible (Z is irreducible by hypothesis, and $\mathbb{A}^3$ trivially is). Both have the **same** dimension $3k - 2$, by Theorem 16 and Lemma 15. Containment would therefore force equality. This is refuted: $Z \times \mathbb{A}^3$ is disjoint from $\pi^{-1}(U_{k-1})$, since $\overline{w}_{k-1}$ vanishes identically on $Z \subset V(I_{k-1})$; but Y_2 meets $\pi^{-1}(U_{k-1})$, by Lemma 15. □

Theorem 20 (component count)

For every $k \geq 2$: $V(I_k)$ has exactly $k - 1$ irreducible components, each of dimension $3k - 2$. The component indexed $j = 2, \ldots, k$ involves exactly the first j coordinate blocks and nothing later.

Proof of Theorem 20. Induction on k. **Base case $k = 2$**: the rank-≤ 1 locus of a generic 2×3 matrix is irreducible – the affine cone over the Segre embedding $\mathbb{P}^1 \times \mathbb{P}^2 \hookrightarrow \mathbb{P}^5$, equivalently the Motzkin-Taussky [3] irreducibility of the 2×2 commuting variety – so exactly 1 component, matching $k - 1 = 1$. **Step**: $V(I_k) = Y_1 \cup Y_2$ as in Theorem 16's proof; by the inductive hypothesis Y_1 has $k - 2$ irreducible components, each of dimension $3k - 2$ (Theorem 16), none contained in another; Proposition 18 and Proposition 19 rule out containment between Y_2 and any of them or vice versa. A finite union of irreducible closed sets with no containments is irredundant, giving exactly $(k - 2) + 1 = k - 1$ components. □

This gives I_k's minimal primes and their count, but not yet that I_k is radical (so not yet that these are literally the components of a **primary** decomposition of I_k itself, as opposed to of $\sqrt{I_k}$) – that is closed in §6, after Cohen-Macaulayness is available.

Definition 21 (the components P_j)

For $k \geq 2$ and $j = 2, \ldots, k$, write P_j for the vanishing ideal of the component of $V(I_k)$ indexed j in Theorem 20 – the one involving exactly the first j coordinate blocks. Explicitly, recursively in j:

- **Base case, $j = k$**: P_k is Y_2's own defining ideal at parameter k (Lemma 12).
- **Recursive case, $j < k$**: P_j is that same ideal computed at parameter j instead – i.e. Y_2's defining ideal when the construction is run with $k = j$ – viewed inside R_k by leaving $\overline{v}_{j+1}, \ldots, \overline{v}_k$ free.
- **Consequence**: P_j does not depend on which $k \geq j$ it is viewed inside of – it is the same ideal, just embedded in a ring with more or fewer free variables.

♣

Example 22 (a suitable point of Y_2, concretely, at $k = 3$)

Rather than only the defining ideal (Example 24 below), a single explicit point: take $\overline{v}_1 = (1, 0, 0)$, $\overline{v}_2 = (0, 1, 0)$ (so $\overline{v}_1 \in U_2$, Lemma 6), giving $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2 = (0, 0, 1)$ and $\tau(\overline{w}_2) = (0, 0, 2)$. Taking $\lambda = 1$ in Definition 11's Φ, i.e. $\overline{v}_3 := \tau(\overline{w}_2) = (0, 0, 2)$, gives the point

$$(\overline{v}_1, \overline{v}_2, \overline{v}_3) = ((1, 0, 0), (0, 1, 0), (0, 0, 2)) \in Y_2 \subset \mathbb{A}^9. \tag{27}$$

Directly: $\overline{w}_3 = \tau(\overline{w}_2) \times \overline{v}_3 = (0, 0, 2) \times (0, 0, 2) = 0$ (a vector crossed with itself), confirming this point lies in $V(I_3)$, as any point of the form $\Phi(p, \lambda)$ must (Definition 11). This is also, concretely, a point of the largest-dimension inherited case: it is **not** a rank-2 witness in Lemma 35's sense (that witness uses $\lambda = 0$, not $\lambda = 1$) – it illustrates Y_2 itself, not the radicality argument's own witness points.

◇

Proposition 23 (rational parametrization of each component)

For $k \geq 2$ and $j = 2, \ldots, k$, the component of $V(I_k)$ indexed j (Theorem 20) is a **rational** variety: it is the closure of the image of the polynomial map

$$\mathbb{A}^{3(j-1)} \times \mathbb{A}^1 \times \mathbb{A}^{3(k-j)} \to \mathbb{A}^{3k}, \quad (p, \lambda, q) \mapsto \big(\Phi_j(p, \lambda), q\big), \tag{28}$$

where Φ_j is Φ (Definition 11) run at parameter j in Definition 21′ sense (at $j = 2$, $\Phi_2(p, \lambda) := (p, \lambda p)$, matching Theorem 4's own base-case convention), and q fills the remaining free blocks $\overline{v}_{j+1}, \ldots, \overline{v}_k$. This map is moreover **birational** onto the component: it is injective over $U_{j-1} \times \mathbb{A}^1 \times \mathbb{A}^{3(k-j)}$, by the same argument as Lemma 15, applied at parameter j.

♠

Proof of Proposition 23. By Definition 21, P_j's defining ideal is Y_2's own defining ideal run at parameter j – i.e. the component itself **is** Y_2 at parameter j, crossed with the free blocks $\overline{v}_{j+1}, \ldots, \overline{v}_k$. Lemma 15's injectivity argument (over U_{j-1}, via Lemma 6 at level $j - 1$) applies verbatim at parameter j, and appending the free blocks q unchanged does not affect injectivity. □

Remark 24 (a reducible variety is not itself "rational" – component-by-component is what Proposition 23 answers)

$Y_1 := V(I_{k-1}) \times \mathbb{A}^3$ (Definition 11) is **not** irreducible for $k \geq 4$ – it is a union of $k - 2$ of the components above (Theorem 20's inductive step) – so a single rational parametrization **of Y_1 as one object** is not meaningful in the usual sense (rationality is a property of irreducible varieties). What **is** meaningful, and what Proposition 23 supplies, is a rational parametrization of each of Y_1's irreducible pieces individually – exactly the components inherited from $V(I_{k-1})$.

◇

Example 25 (the components, worked out at $k = 3$)

Uniformly by k, since the components' **count** and **block pattern** are already known (Theorem 20, Definition 21) and only their explicit generators change:

- **At $k = 2$**: I_2 is already prime (Theorem 20′s base case), so $V(I_2)$ has exactly $k - 1 = 1$ component – itself.
- **At $k = 3$**: exactly $k - 1 = 2$ components, both computed directly (Gröbner bases, not inferred from the pattern alone), matching $I_3 = P_2 \cap P_3$:
  - P_2, the "Y_1" piece (block pattern $\{1, 2\}$, matching that the length-2 sub-bracket already vanishes with $\overline{v}_3$ free) – generated by exactly I_2's three generators from the first example, now inside R_3 with $\overline{v}_3$ simply not appearing:
  $$v_{1,2}v_{2,3} - v_{1,3}v_{2,2}, \quad v_{1,3}v_{2,1} - v_{1,1}v_{2,3}, \quad v_{1,1}v_{2,2} - v_{1,2}v_{2,1}. \tag{29}$$
  - P_3, the "Y_2" piece (block pattern $\{1, 2, 3\}$) – exactly the ideal already computed in the Y_2-defining-ideal example above:
  $$2v_{2,2}v_{3,1} + 2v_{2,1}v_{3,2} + v_{2,3}v_{3,3}, \quad 2v_{1,2}v_{3,1} + 2v_{1,1}v_{3,2} + v_{1,3}v_{3,3}, \tag{30}$$
  $$2v_{1,2}v_{2,1}v_{3,2} - 2v_{1,1}v_{2,2}v_{3,2} - v_{1,3}v_{2,2}v_{3,3} + v_{1,2}v_{2,3}v_{3,3}. \tag{31}$$
- **At $k = 4$**: exactly $k - 1 = 3$ components, block patterns $\{1, 2\}$, $\{1, 2, 3\}$, $\{1, 2, 3, 4\}$:
  - The first two, P_2 and P_3, are inherited **verbatim** from $k = 3$ above, with $\overline{v}_4$ simply left free – not recomputed, per Definition 21′ own "does not depend on which k" clause.
  - The third, P_4, is the new "Y_2" stratum – exactly the $k = 4$ row of the earlier Y_2-defining-ideal example (generators omitted here to avoid repeating that display; same growth pattern, quartic with up to 8 terms per generator).

6. Cohen-Macaulayness

Proposition 26 (Eagon-Northcott / Hilbert-Burch, applied)

Let M be a 2×3 matrix with entries in a polynomial ring R over a field (arbitrary entries, not necessarily independent indeterminates), and J the ideal of its 2×2 minors. If J has the **expected** codimension $3 - 2 + 1 = 2$, then J is a perfect ideal of grade 2 – $\mathrm{pd}(R/J) = 2$, hence R/J is Cohen-Macaulay.

Proof of Proposition 26. Bruns and Vetter [1], Theorem (2.7), p. 13, gives exactly this for an **arbitrary** matrix, as needed since one row of our 2×3 matrix $(\tau(\overline{w}_{k-1}))$ is a fixed polynomial vector, not a row of independent indeterminates. (The original construction, Eagon and Northcott [2], is superseded here by Bruns-Vetter's more general treatment – confirmed closed-access via Unpaywall, not used directly.) □

Remark 27 (why the generator count above wasn't proven this way)

It is tempting to expect Proposition 26 to also hand over the generator count for free via Bruns-Vetter's Theorem (16.36), p. 217 ("The Theorem of Hilbert-Burch") – but that theorem, as actually stated, gives only $I = a \cdot I_m(f)$ for a nonzerodivisor a, not a generator count on its own: at expected codimension, an **arbitrary** 2×3 matrix's minors can be perfect and Cohen-Macaulay while one minor vanishes identically or is a scalar combination of the others (concretely: $\begin{pmatrix} x & y & 0 \\ 0 & 0 & 1 \end{pmatrix}$ over $\mathbb{F}[x, y]$ has minors $0, x, y$, height 2 as expected, $J = (x, y)$ perfect and Cohen-Macaulay with only **2** generators). This is exactly why Theorem 9 was proven directly from linear independence instead of cited from this section's machinery.

Theorem 28 (R_k/I_k is Cohen-Macaulay)

For every $k \geq 2$: R_k/I_k is Cohen-Macaulay.

Proof of Theorem 28. By Theorem 16, I_k has codimension 2, the expected value for the 2×3 matrix of Theorem 4 presenting it. Proposition 26 applies directly. (This uses Theorem 16 only – not Theorem 9′s generator count, not Theorem 20′s component count – genuinely independent of both.) □

7. Radicality

Cohen-Macaulayness and the explicit component structure of §5 combine, via Serre's criterion, to give radicality – which then closes both the generator-count (§4) / Cohen-Macaulay (§6) story (making I_k itself, not just $\sqrt{I_k}$, the object with those properties) and upgrades §5′s component count to a genuine primary decomposition.

Lemma 29 (reduction via Serre)

Recall Serre's condition S_n: a Noetherian ring A satisfies S_n if $\operatorname{depth}(A_P) \geq \min(n, \dim A_P)$ for every prime $P \subset A$ (Cohen-Macaulay means $\operatorname{depth}(A_P) = \dim A_P$ at **every** prime – the extremal case, $n = \infty$ in effect). Cohen-Macaulay rings satisfy S_n for every n: $\operatorname{depth}(A_P) = \dim A_P \geq \min(n, \dim A_P)$ trivially, whatever n is – this needs nothing beyond the two definitions just given, standard for Serre's conditions in general (see e.g. Bruns and Vetter [1] or Eisenbud's *Commutative Algebra with a View Toward Algebraic Geometry*). In particular S_1. So, given Theorem 28, radicality of I_k is equivalent to R_0: regularity of R_k/I_k at each of its minimal primes.

Lemma 30 (Jacobian criterion: the rank bound)

Let $S = \mathbb{F}[x_1, \ldots, x_n]$, $I = (f_1, \ldots, f_m)$, $A = S/I$ equidimensional of dimension $n - c$, and $x \in V(I)$ an $\mathbb{F}$-rational point with Jacobian $\operatorname{Jac}(x) \in \mathbb{F}^{m \times n}$. Then rank $\operatorname{Jac}(x) \leq c$.

Proof of Lemma 30. $\dim_{\mathbb{F}} \mathfrak{m}/\mathfrak{m}^2 = n - \operatorname{rank}\ \operatorname{Jac}(x)$ always (linear algebra: the image of I in the cotangent space $\mathfrak{n}/\mathfrak{n}^2 \cong \mathbb{F}^n$ is exactly the span of the linear parts of generators of I, of dimension rank $\operatorname{Jac}(x)$), and $\dim_{\mathbb{F}} \mathfrak{m}/\mathfrak{m}^2 \geq \dim A_{\mathfrak{m}}$ in any Noetherian local ring, so it suffices that $\dim A_{\mathfrak{m}} = n - c$ – which Lemma 31 supplies. □

Lemma 31 (Jacobian criterion: regularity at rank $= c$)

With hypotheses as in Lemma 30: $\dim A_{\mathfrak{m}} = n - c$. Consequently, if rank $\operatorname{Jac}(x) = c$, then $A_{\mathfrak{m}}$ is regular of dimension $n - c$; in particular x lies on a **unique** component of $V(I)$, whose local ring there is a field.

Proof of Lemma 31. $\dim A_{\mathfrak{m}} = \max\{\dim (A/P)_{\mathfrak{m}} : P \in \operatorname{Min}(A), P \subseteq \mathfrak{m}\}$, and each $A/P = S/\tilde{P}$ is an affine domain over a field, for which $\dim \left(S/\tilde{P}\right)_{\mathfrak{M}} = \dim S/\tilde{P}$ at **every** maximal ideal $\mathfrak{M}$ – affine domains over a field are catenary and equicodimensional (Eisenbud, *Commutative Algebra*, Cor. 13.4; equivalently the Stacks Project, tag 00OS [4] – a dimension formula for finitely generated field extensions, holding over an arbitrary field, not needing a perfect-field hypothesis by accident). By equidimensionality every such $\dim S/\tilde{P}$ equals $n - c$. If rank $\operatorname{Jac}(x) = c$ then $\dim_{\mathbb{F}} \mathfrak{m}/\mathfrak{m}^2 = n - c = \dim A_{\mathfrak{m}}$ by Lemma 30, the definition of regularity. A regular local ring is a domain, so $\mathfrak{m}$ contains a unique minimal prime of A, and the local ring at that prime is a further localization of a regular local ring at its own maximal ideal in dimension 0, hence a field. □

Lemma 32 (the Jacobian's own recursion)

Write J_k for the $3 \times 3k$ Jacobian of $\overline{w}_k$, $[u]_\times$ for the skew matrix with $[u]_\times v = u \times v$, and T for the matrix of τ. Then $J_2 = \left[-[\overline{v}_2]_\times \;\middle|\; [\overline{v}_1]_\times\right]$, and for $k \geq 3$,

$$J_k = \left[-[\overline{v}_k]_\times T J_{k-1} \;\middle|\; [\tau(\overline{w}_{k-1})]_\times\right]. \tag{32}$$

Proof of Lemma 32. For the second block, $\overline{w}_k = \tau(\overline{w}_{k-1}) \times \overline{v}_k = [\tau(\overline{w}_{k-1})]_\times \overline{v}_k$ is linear in $\overline{v}_k$ with coefficient matrix free of $\overline{v}_k$. For the first, differentiating $\overline{w}_k = \tau(\overline{w}_{k-1}) \times \overline{v}_k$ with respect to any variable t among $\overline{v}_1, \ldots, \overline{v}_{k-1}$ gives $\partial_t \overline{w}_k = (T\partial_t \overline{w}_{k-1}) \times \overline{v}_k = -[\overline{v}_k]_\times T\partial_t \overline{w}_{k-1}$, using $a \times b = -[b]_\times a$; assembling columns gives the stated product. □

Example 33 (J_3, explicitly)

With variables ordered $v_{1,1}, v_{1,2}, v_{1,3}, v_{2,1}, v_{2,2}, v_{2,3}, v_{3,1}, v_{3,2}, v_{3,3}$, the 3×9 Jacobian of $\overline{w}_3$ (the three generators of I_3 from the earlier example), column-blocked to match Lemma 32's formula $J_3 = \left[-[\overline{v}_3]_\times T J_2 \;\middle|\; [\tau(\overline{w}_2)]_\times\right]$ – shown here as the two blocks separately, since side by side the full 3×9 matrix does not fit the page width. First, $-[\overline{v}_3]_\times T J_2$ (3×6):

$$\begin{pmatrix} -2v_{2,2}v_{3,2} & 2v_{2,1}v_{3,2} + v_{2,3}v_{3,3} & -v_{2,2}v_{3,3} & 2v_{1,2}v_{3,2} & -2v_{1,1}v_{3,2} - v_{1,3}v_{3,3} & v_{1,2}v_{3,3} \\ 2v_{2,2}v_{3,1} + v_{2,3}v_{3,3} & -2v_{2,1}v_{3,1} & -v_{2,1}v_{3,3} & -2v_{1,2}v_{3,1} - v_{1,3}v_{3,3} & 2v_{1,1}v_{3,1} & v_{1,1}v_{3,3} \\ -v_{2,3}v_{3,2} & -v_{2,3}v_{3,1} & v_{2,2}v_{3,1} + v_{2,1}v_{3,2} & v_{1,3}v_{3,2} & v_{1,3}v_{3,1} & -v_{1,2}v_{3,1} - v_{1,1}v_{3,2} \end{pmatrix} \tag{33}$$

and then $[\tau(\overline{w}_2)]_\times$ (3×3, the skew matrix of $\tau(\overline{w}_2)$'s three coordinates from the earlier matrix example):

$$\begin{pmatrix} 0 & 2v_{1,2}v_{2,1} - 2v_{1,1}v_{2,2} & -v_{1,3}v_{2,2} + v_{1,2}v_{2,3} \\ -2v_{1,2}v_{2,1} + 2v_{1,1}v_{2,2} & 0 & -v_{1,3}v_{2,1} + v_{1,1}v_{2,3} \\ v_{1,3}v_{2,2} - v_{1,2}v_{2,3} & v_{1,3}v_{2,1} - v_{1,1}v_{2,3} & 0 \end{pmatrix} \tag{34}$$

Both blocks computed directly, not merely asserted from the recursive formula. ◇

Remark 34 (a false shortcut, caught and corrected)

A naive reading of Lemma 32 suggests appending $\overline{v}_k = 0$ preserves rank on an inherited component – this is false: the old-variable block is left-multiplied by $-[\overline{v}_k]_\times T$, not merely padded with zero rows. At the first inherited case ($k = 2 \to 3$), appending $\overline{v}_3 = 0$ actually drops the rank to 0. The correct witness (Lemma 36 below) instead picks $\overline{v}_k = e_i$ for a standard basis vector **not** lying in $T \cdot \operatorname{im} J_{k-1}(q)$ – a genuine choice, not a formality. ◇

Lemma 35 (rank-2 witness on the new component)

For every $k \geq 2$, there is a point $x = (p, 0)$, p the Lemma 6 non-degeneracy witness at level $k-1$, with $x \in V(I_k)$ and rank $J_{k(x)} = 2$. ♡

Proof of Lemma 35. By Lemma 32, $J_{k(p,0)} = \left[0 \;\middle|\; [\tau(\overline{w}_{k-1}(p))]_\times\right]$ since $-[0]_\times T J_{k-1}(p) = 0$; and $\tau(\overline{w}_{k-1}(p)) \neq 0$ since $\overline{w}_{k-1}(p) \neq 0$ (Lemma 6) and τ is invertible. For $u \neq 0$, $\ker [u]_\times = \{v : u \times v = 0\} = \mathbb{F}u$ (a cross product vanishes exactly on linearly dependent pairs), so $[\tau(\overline{w}_{k-1}(p))]_\times$ has rank 2, giving rank $J_{k(p,0)} = 2$. □

Lemma 36 (rank-2 witness on an inherited component)

For every $k \geq 3$ and every component Z of $V(I_{k-1})$ with a rank-2 witness $q \in Z$ (Lemma 35, or this lemma applied at $k-1$), there is $x = (q, e_i) \in Z \times \mathbb{A}^3 \subset V(I_k)$ with rank $J_{k(x)} = 2$. ♡

Proof of Lemma 36. Let $W := T \cdot \operatorname{im} J_{k-1}(q)$, a 2-dimensional subspace of $\mathbb{F}^3$ since T is invertible and rank $J_{k-1}(q) = 2$. Since e_1, e_2, e_3 span $\mathbb{F}^3$ and $W \neq \mathbb{F}^3$, some $e_i \notin W$; fix such an i. Since $q \in V(I_{k-1})$, $\overline{w}_{k-1}(q) = 0$, so Lemma 32 gives $J_{k(q,e_i)} = \left[-[e_i]_\times T J_{k-1}(q) \;\middle|\; 0\right]$, of rank $\dim\left([e_i]_\times W\right)$ since $T J_{k-1}(q)$ has image exactly W. As $\ker [e_i]_\times = \mathbb{F}e_i$ and $W \cap \mathbb{F}e_i = 0$ (else $e_i \in W$), $[e_i]_\times$ is injective on W, so $\dim\left([e_i]_\times W\right) = 2$. □

Theorem 37 (I_k is radical)

For every $k \geq 2$: I_k is radical. ♡

Proof of Theorem 37. By induction on k (mirroring Theorem 20's own induction), Lemma 35 and Lemma 36 exhibit a rank-2-Jacobian point on **every** one of the $k-1$ components of Theorem 20. Lemma 31 then gives R_0 at every one (and, by its "unique component" clause, confirms each witness lies only on its intended component – an independent check on Theorem 20's count, not merely assumed); Lemma 29 gives S_1, using Theorem 28's Cohen-Macaulayness. Serre's criterion gives radicality. □

Verified independently in Sage at every $k = 2, \ldots, 6$: the Jacobian recursion (Lemma 32) checked as a polynomial identity, not just at points, and each witness confirmed to lie on **exactly** its intended component via an independently-computed primary decomposition (*code/verify_jacobian_rank_all_components.sage*).

8. The Primary Decomposition Theorem

Every ingredient is now in hand for the fully general statement – component count (§5) and radicality (§7) together, neither sufficient alone:

Theorem 38 (Primary Decomposition Theorem)

For every $k \geq 2$, over any field of characteristic $\neq 2$: I_k has exactly $k-1$ irreducible, radical, primary components, namely $P_2, \ldots, P_k$ of Definition 21, and

$$I_k = P_2 \cap \cdots \cap P_k \tag{35}$$

is an irredundant primary decomposition into $k-1$ primes.

Proof of Theorem 38. Spelled out in full, combining the two independent halves proven above rather than left as a one-line conjunction. By Theorem 37, I_k is radical, i.e. $I_k = \sqrt{I_k}$. A standard fact about radical ideals in a Noetherian ring: a radical ideal equals the intersection of its own minimal primes, $I_k = \cap_{P \in \operatorname{Min}(I_k)} P$ – since $\sqrt{I_k}$ is, by definition, the intersection of **all** primes containing I_k, and the minimal ones already suffice (every non-minimal prime containing I_k contains some minimal one, hence is redundant in the intersection). Theorem 20 identifies $\operatorname{Min}(I_k)$ explicitly: exactly $k-1$ primes, one per block pattern $j = 2, \ldots, k$, named $P_2, \ldots, P_k$ by Definition 21. Substituting gives the displayed equation. Each P_j, being prime, is trivially its own primary-decomposition component (P_j-primary with associated prime P_j itself) – no separate argument for "primary" beyond "prime" is needed, and primality of each P_j is exactly Lemma 12 (applied at the appropriate parameter, via Theorem 37's own induction). Irredundancy – no P_j can be omitted, i.e. no P_j contains $\cap_{i \neq j} P_i$ – is exactly Theorem 20's own no-containment conclusion (Proposition 17 and Proposition 18, applied at every level of that theorem's induction), not a fact needing any further proof here. □

Remark 39 (summary)

Theorem 16 (codimension 2), Theorem 9 (3 minimal generators), Theorem 28 (Cohen-Macaulay), and Theorem 37 (radical) together say exactly what a single bundled "I_k has codimension 2, is radical, has exactly 3 minimal generators, and is Cohen-Macaulay" statement would – now as four independently-proven results, each established at the point in this article where its actual dependencies (and no others) were available, rather than one claim proven in parts across several sections.

9. Summary

For every $k \geq 2$, over any field $\mathbb{F}$ of characteristic $\neq 2$:

- I_k has codimension 2 (Theorem 16), presented as the ideal of 2×2 minors of an explicit 2×3 matrix (Theorem 4) – a genuine determinantal ideal, not merely determinantal-looking.
- I_k has exactly 3 minimal generators (Theorem 9), proven directly from linear independence, independent of the determinantal presentation.
- R_k/I_k is Cohen-Macaulay (Theorem 28), from the determinantal presentation and classical Eagon-Northcott/Hilbert-Burch theory (Proposition 26).
- I_k is radical (Theorem 37), via Serre's criterion: Cohen-Macaulayness gives S_1 (Lemma 29), and an explicit rank-2 Jacobian witness on every component gives R_0 (Lemma 35, Lemma 36).
- $V(I_k)$ has exactly $k-1$ irreducible components (Theorem 20), each rational – in fact birational to an explicit affine parameter space (Proposition 23) – and these assemble, via the two facts above, into the irredundant primary decomposition of the Primary Decomposition Theorem (Theorem 38): $I_k = P_2 \cap \cdots \cap P_k$.

Nothing above is conjectural or checked only at small k: every statement is proven for **every** $k \geq 2$, over an **arbitrary** field of the stated characteristic. The one running computational thread – Sage verification of the determinantal identity, the Jacobian recursion, and the witness points, at $k = 2, \ldots, 6$ – corroborates the general proofs; it is not a substitute for them anywhere in this article.

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